

Zero Before Separation in a Primitive Möbius Tail: Provenance-Weighted Closure and the Principal Row-Mode Boundary

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Abstract

TPC-24 proves pointwise arbitrary-logarithmic cancellation for the physical additive zero mode of a calibrated one-sided divisor shell, but its abstract packet theorem retains an additional fixed-projective- mass hypothesis. We return here to the actual primitive opened- d packet before smooth row-orbit separation. Its row coefficient has the provenance $\mu(d)(\log \ell)$, up to fixed bounded cutoffs, where $m = \ell d$, $\ell \asymp L$ is prime, $d \asymp D$, and $Q = LD$. Extracting Poisson frequency zero first turns the original row-dependent smooth orbit weight into a uniformly bounded integral. Chebyshev's estimate then gives

$$\sum_{\substack{\ell \asymp L \\ \ell \text{ prime}}} (\log \ell) \sum_{d \asymp D} |\mu(d)| \ll LD = Q,$$

so the physical zero-mode row-pair mass is $O(Q^2)$, without a projective decomposition and without an X^ε loss.

Combining this zero-before-separation principle with the pointwise shell kernel of TPC-24 yields, for $S = X^{s+o(1)}$, every $0 < \kappa < s$, and every fixed $A > 0$,

$$|Z_{S,T}^0| \ll JQ^2 \{(\log X)^{-A} + X^{-s+\kappa+o(1)}\}.$$

Consequently the fixed-projective-mass condition in TPC-24 is automatic for the actual opened- d one-sided packet. Under the remaining inherited row, no-wrap, Fourier-tail, and conductor interfaces, and when

$$\eta_{\text{asym}}(\beta; s, t) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 2s - 2t}{2}, \frac{3 - \beta - s - 5t}{4} \right\} > 0$$

with fixed margin, the calibrated mixed increment has arbitrary logarithmic saving for a general nonzero-mean fixed smooth test within the factorable provenance class; no mean-zero hypothesis on the fixed physical test is needed. This closes the mass gate for that specific physical packet, not for arbitrary abstract row arrays.

We also prove that the distinction is necessary. Soft row L^2 energy, a divisor-independent off-diagonal mask, and an entrywise kernel bound smaller than every fixed logarithmic power do not imply a packet saving. A dense constant row mode saturates the natural scale while obeying all three interfaces. The same coherent direction occurs in the exact Möbius algebra: for $H = 1$, $R = S = 2$, $T = 3$, and prime rows larger than 3, the raw kernel equals $\kappa_0 = ((\log 2 - 2) \log 3)/6$ on every distinct row pair. After off-diagonal masking its matrix is

$$\kappa_0(\mathbf{1}\mathbf{1}^\top - I).$$

Thus a direct Hilbert–Schmidt upgrade has no dimension gain. Finally, in a prime-row no-long-wrap model we decompose the whole kernel into residue-class constant spaces modulo

the shared factor g . This gives an exact congruence skeleton and a deflated operator bound, while also showing precisely which extra row-balance statistics a future both-new-shell argument would have to control. Nothing here proves a Hardy–Littlewood asymptotic, a twin-prime lower bound, or a general breach of the sieve parity barrier.

Keywords: primitive Möbius tail; physical zero mode; row provenance; principal row mode; congruence skeleton; projective mass; sieve parity.

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1 Introduction

The primitive Möbius-tail program of TPC-18–24 reorganizes a fixed-shift prime-pair residual into dyadic row packets $m = \ell d$, with a large prime source ℓ and a shorter squarefree factor d [2–8]. TPC-24 reaches a genuine one-sided divisor shell

$$u \leq S < v \leq T$$

in which the new endpoint T may lie beyond the Poisson length J . Its nonzero additive frequencies are controlled by an asymmetric exact-conductor rectangle. The physical additive zero frequency is treated separately through the shared divisor $g = (u, v) \mid m - n$. That argument gives, uniformly on every retained off-diagonal row pair,

$$K_{S,T}^0(m, n) \ll_{A,\kappa} (\log X)^{-A} + X^{-s+\kappa+o(1)}, \quad S = X^{s+o(1)}, \quad 0 < \kappa < s. \quad (1)$$

The first term comes from cancellation between the two endpoints of a complete Möbius shell; the second comes from large shared factors of the nonzero row determinant.

The remaining issue is not hidden in (1). TPC-24 states its separated row interface with

$$\|x\|_2^2 + \|y\|_2^2 \ll QX^\varepsilon, \quad \sum_{m,n} |x_m y_n \mathbf{m}(m, n)| \ll Q^2 X^\varepsilon.$$

Multiplying the first term of (1) by X^ε does not produce decay for fixed A and ε . Accordingly TPC-24 keeps a separate fixed-polylogarithmic projective- mass hypothesis, or else assumes that the test function has zero physical mean. Its conclusion section isolates two possible next steps: obtain a row-averaged operator estimate, or recover a fixed-polylogarithmic projective decomposition from the actual packet.

The main point of the present paper is that the physical packet should not be separated before frequency zero is extracted. Before that separation, the opened- d quadratic form has the explicit one-row provenance

$$w_{\ell,d} = \mu(d)(\log \ell) \times \text{a bounded dyadic factor}. \quad (2)$$

For fixed rows, Poisson frequency zero merely integrates the uniformly bounded orbit weight. The resulting two-row scalar is still bounded, and Chebyshev’s estimate gives

$$\sum_{\ell,d} |w_{\ell,d}| \ll LD = Q. \quad (3)$$

Thus the physical zero-frequency pair mass is $O(Q^2)$. This is a provenance statement about the actual TPC-18 packet, not a consequence of the abstract soft L^2 interface.

1.1 Main results

The paper has three logically distinct conclusions.

First, the zero-before-separation theorem inserts (3) directly into (1). For one actual opened- d packet it gives

$$|\mathcal{Z}_{S,T}^0| \ll JQ^2 \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} \right\}. \quad (4)$$

Fixed bounded-overlap dyadic, content, and residue reassembly costs at most a fixed logarithmic power. The calibration exponent A is then chosen after that cost. Consequently the extra projective-mass hypothesis of TPC-24 is verified, rather than assumed, for the actual one-sided packet.

Second, combine (4) with the nonzero-conductor theorem of TPC-24. If

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad S = X^{s+o(1)}, \quad T = X^{t+o(1)}$$

and

$$\eta_{\text{asym}}(\beta; s, t) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 2s - 2t}{2}, \frac{3 - \beta - s - 5t}{4} \right\} > 0 \quad (5)$$

with fixed margin, then the actual calibrated mixed increment has arbitrary logarithmic saving under the remaining inherited interfaces. This conclusion allows a general nonzero-mean fixed smooth test within the factorable provenance class. At $S = R = X^{1/2-\delta+o(1)}$ and $T = RX^{\xi+o(1)}$, it retains the strict window $R < J < T$ already identified by TPC-24.

Third, we prove that this positive result cannot be promoted to arbitrary arrays satisfying only the soft row interfaces. The shared factor $g = 1$ has no row oscillation. A constant off-diagonal kernel whose entries are $\exp(-\sqrt{\log X})$ is smaller than every fixed negative power of $\log X$, yet suitable $X^{o(1)}$ -sized row weights make its bilinear form saturate Q^2 . This abstract countermodel is reflected by a finite exact Möbius kernel:

$$H = 1, \quad R = S = 2, \quad T = 3 \quad \implies \quad K_{2,3}^0(m, n) = \frac{(\log 2 - 2) \log 3}{6}$$

for every pair of distinct prime rows larger than 3. Hence the operator norm and Hilbert–Schmidt norm are both of order the number of rows; a direct Hilbert–Schmidt argument has no dimensional gain.

We finally identify the complete low-row-frequency geometry in a prime-row no-long-wrap model. The g -layer is constant on residue classes modulo g , so the correlated part of the kernel lies in the span of residue-class indicators. Projecting those spaces away gives an exact deflated operator estimate. This is a concrete route for a future both-new-shell problem, but the generic Schur mask and the general d -block geometry must then be retained explicitly.

1.2 Logical scope

The positive closure in this paper concerns the physical provenance class (2). It does not apply to an arbitrary uniformly smooth family F_{α_1, α_2} with arbitrary coefficients z_{α_1, α_2} , nor does it prove that every energy-bounded row vector is balanced against the principal row mode. Conversely, the energy-only countermodel is a logical non-implication, not a lower bound for the wide Möbius shells in the target exponent region.

Only one mixed component $P_{m,S}(P_{n,T} - P_{n,S})$, and its transpose, is closed here. The both-new square, the ultra-long complement fibers, the full residual reassembly, positivity, and a prime-pair main term remain separate stages. In particular no assertion below is a Hardy–Littlewood asymptotic, a twin-prime theorem, or a general violation of the sieve parity barrier.

2 The actual opened-row provenance

The logarithmic zero-mode loss in the abstract separated interface is not present in the packet from which that interface was obtained. The point is elementary but order-sensitive: the original opened row has an explicit prime–Möbius weight whose ℓ^1 -mass is $O(Q)$. We record the provenance before taking any Mellin–Fourier separation.

Fix $h \in \mathbb{Z} \setminus \{0\}$, put

$$H = \text{rad } |h|, \quad H_0 = |h|, \quad (6)$$

and fix one dyadic opened- d slice. Here H controls primitivity, while the fixed period H_0 retains the full residue masks inherited from TPC-18. Put

$$R = X^{1/2-\delta+o(1)}, \quad V = R^{1/2+o(1)}, \quad Q = LD, \quad J = K/D, \quad QJ \asymp X, \quad (7)$$

and assume

$$L > \max\{2R, 2|h|\}. \quad (8)$$

Its rows are

$$\begin{aligned} \alpha &= (\ell, d), & \ell &\in [L/2, 2L] \text{ prime}, & m_\alpha &= \ell d, \\ d &\in [D, 2D] \cap [1, V], & d &\text{squarefree}, & (d, H) &= 1. \end{aligned} \quad (9)$$

The size separation makes the integer row m_α determine its label uniquely. A typical one-row coefficient is

$$\gamma_\alpha = \mu(d)(\log \ell)\omega_D(d)\psi_L(\ell/L)\zeta_\alpha, \quad |\zeta_\alpha| \leq C, \quad (10)$$

where ω_D and ψ_L range over fixed bounded smooth families. The harmless factor ζ_α permits fixed residue characters and unit-modulus phases. The two sides of a bilinear packet may carry different coefficients $\gamma_\alpha^{(1)}$ and $\gamma_\alpha^{(2)}$ of this form.

The orbit amplitude before separation is a product of fixed smooth factors such as

$$W(m_\alpha j/X) \Psi(dj/K) \quad (11)$$

and a mask $\xi(d, j)$ depending on $(d, j) \bmod H_0$. After the quadratic expansion, the generic Schur mask $\mathbf{m}(\alpha_1, \alpha_2)$ is imposed on the row pair. We use only

$$|\mathbf{m}(\alpha_1, \alpha_2)| \leq 1, \quad \mathbf{m}(\alpha_1, \alpha_2) = 0 \quad \text{whenever } m_{\alpha_1} = m_{\alpha_2}. \quad (12)$$

In particular, the mask never increases an absolute row-pair mass.

2.1 Chebyshev row mass

Lemma 2.1 (Actual one-row ℓ^1 -mass). *For each fixed opened slice and each of the two row factors,*

$$\boxed{\sum_{\alpha} |\gamma_\alpha^{(i)}| \ll Q, \quad i = 1, 2.} \quad (13)$$

The implied constant depends only on the fixed smooth and residue families. No prime number theorem in an arithmetic progression is used.

Proof. Chebyshev’s estimate and boundedness of the source cutoff give

$$\sum_{\substack{\ell \in [L/2, 2L] \\ \ell \text{ prime}}} (\log \ell) |\psi_L(\ell/L)| \ll L. \quad (14)$$

Also

$$\sum_{d \in [D, 2D]} |\mu(d)\omega_D(d)| \ll D. \quad (15)$$

Multiplication and $LD = Q$ prove (13). Notice that the factor $\log \ell$ has already been summed by (14); replacing it by a generic X^ε bound at this point would lose genuine information. $\square$

Lemma 2.2 (Actual one-row energy). *For each row factor in (10),*

$$\|\gamma^{(i)}\|_\infty \ll \log(2L), \quad \|\gamma^{(i)}\|_2^2 \ll Q \log(2L), \quad i = 1, 2. \quad (16)$$

Proof. The first estimate is immediate. For the second, Chebyshev's estimate and $(\log \ell)^2 \leq \log(2L) \log \ell$ give

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}|^2 \ll D \sum_{\substack{\ell \asymp L \\ \ell \text{ prime}}} (\log \ell)^2 \ll DL \log(2L) = Q \log(2L).$$

□

Corollary 2.3 (Actual pair mass). *Uniformly for every bounded row-pair mask,*

$$\sum_{\alpha_1, \alpha_2} |\gamma_{\alpha_1}^{(1)} \gamma_{\alpha_2}^{(2)} \mathbf{m}(\alpha_1, \alpha_2)| \ll Q^2. \quad (17)$$

The same conclusion holds after multiplication by any uniformly bounded scalar depending on the row pair.

Proof. Delete the mask in absolute value and apply lemma 2.1 to the two factors. □

2.2 Smooth and finite-residue provenance

There are two different facts here. The first, which is all that the physical zero frequency needs, is that the row-dependent smooth orbit weight has a uniformly bounded integral. Indeed, on (7)–(9),

$$\frac{m_{\alpha} J}{X} \asymp 1, \quad \frac{dJ}{K} \asymp 1. \quad (18)$$

Consequently every product of the factors (11), written as a function of $t = j/J$, belongs to a fixed bounded subset of $C_c^\infty(\mathbb{R})$. In particular, if it is denoted by $F_{\alpha_1, \alpha_2}(t)$, then

$$\sup_{\alpha_1, \alpha_2} \left| \widehat{F}_{\alpha_1, \alpha_2}(0) \right| \leq \sup_{\alpha_1, \alpha_2} \int_{\mathbb{R}} |F_{\alpha_1, \alpha_2}(t)| dt \ll 1. \quad (19)$$

The fixed residue mask introduces no growing complexity. For example, Fourier inversion on $(\mathbb{Z}/H_0\mathbb{Z})^2$ writes

$$\xi(d, j) = \sum_{a, b \bmod H_0} \widehat{\xi}_{H_0}(a, b) e_{H_0}(ad + bj), \quad \sum_{a, b \bmod H_0} |\widehat{\xi}_{H_0}(a, b)| \ll_h 1. \quad (20)$$

Equivalently, one may first split the $O_h(1)$ residue cells. In either description, the physical zero coefficient acquires only a bounded local factor.

For completeness, the usual smooth separation also has no intrinsic X^ε projective loss. If

$$\widetilde{W}(\tau) = \int_0^\infty W(z) z^{-i\tau} \frac{dz}{z}, \quad (21)$$

then $\widetilde{W}$ is Schwartz and Mellin inversion separates $W(mj/X)$ into a row phase and an orbit phase, both of modulus one on the positive support. Applying this to the four smooth factors in a two-row packet gives a continuous projective representation

$$\boldsymbol{\tau} = (t_1, s_1, t_2, s_2), \quad d\nu(\boldsymbol{\tau}) = (2\pi)^{-4} \widetilde{W}(t_1) \widetilde{\Psi}(s_1) \widetilde{W}(t_2) \widetilde{\Psi}(s_2) d\boldsymbol{\tau},$$

with the evident replacement when the two sides carry different fixed smooth functions. The row phases and fixed coefficients are denoted by $x_{\boldsymbol{\tau}}, y_{\boldsymbol{\tau}}$, while all orbit phases are placed in $\Phi_{\boldsymbol{\tau}}$. Thus

$$\gamma_{\alpha_1}^{(1)} \gamma_{\alpha_2}^{(2)} F_{\alpha_1, \alpha_2}(j/J) = \int_{\mathbb{R}^4} x_{\boldsymbol{\tau}}(\alpha_1) y_{\boldsymbol{\tau}}(\alpha_2) \Phi_{\boldsymbol{\tau}}(j/J) d\nu(\boldsymbol{\tau}), \quad (22)$$

after the fixed row coefficients have been included in x and y . For every fixed integer $M \geq 0$, Schwartz decay and [lemma 2.1](#) give

$$\int_{\mathbb{R}^4} (1 + |\boldsymbol{\tau}|)^M \|x_{\boldsymbol{\tau}}\|_1 \|y_{\boldsymbol{\tau}}\|_1 d|\nu|(\boldsymbol{\tau}) \ll_M Q^2. \quad (23)$$

The residue expansion (20) changes only the implied constant. Formula (22) may be replaced by finite Riemann sums and passed to the limit by dominated convergence. Thus there is no hidden infinite projective norm.

The same representation supplies the nonzero-frequency norm interface used later. Unit-modulus Mellin phases do not change (16), and Schwartz decay gives, for every fixed integer $M \geq 0$,

$$\int_{\mathbb{R}^4} (1 + |\boldsymbol{\tau}|)^M \|x_{\boldsymbol{\tau}}\|_2 \|y_{\boldsymbol{\tau}}\|_2 d|\nu|(\boldsymbol{\tau}) \ll_M Q \log(2L). \quad (24)$$

Thus a separated estimate whose constant is polynomial in finitely many orbit seminorms may be integrated term by term. The fixed H_0 -residue expansion costs only $O_h(1)$, and dominated convergence restores the complete smooth packet.

The frequently used statement that one may truncate Mellin variables at X^ε with power accuracy is a convenient coarse interface, not a lower bound for the projective mass. For a requested logarithmic accuracy one may either keep the absolutely convergent integral, as above, or truncate at a power of $\log X$; the omitted tail is then arbitrarily logarithmically small by (23). More importantly, the next section shows that the physical zero frequency should be taken before any such separation.

3 The physical zero frequency before smooth separation

We now apply the provenance of [section 2](#) to the one-sided primitive Möbius shell of TPC-24. Put

$$a(v) = -\mu(v) \log v, \quad (25)$$

and let $b_R(u)$ be the calibrated prefix coefficient. For an off-diagonal row pair $m_1 \neq m_2$, define its physical zero kernel by

$$K_{S,T}^0(m_1, m_2) := \sum_{\substack{u \leq S, S < v \leq T \\ (u, m_1 H) = (v, m_2 H) = 1}} \frac{b_R(u) a(v)}{uv} (u, v) \mathbf{1}_{(u,v) | m_1 - m_2}. \quad (26)$$

The pointwise shell theorem of TPC-24 states that, if $R \leq S < T \leq U$, $S = X^{s+o(1)}$, $T \leq X^{C_T}$, and $0 < \kappa < s$, then for every fixed $A > 0$, uniformly over the retained off-diagonal rows,

$$K_{S,T}^0(m_1, m_2) \ll_{A, \kappa, C_T, h} (\log X)^{-A} + X^{-s+\kappa+o(1)}. \quad (27)$$

Its small-shared-factor term uses cancellation between the two complete endpoints $S/(u, v)$ and $T/(u, v)$; its large-shared-factor term uses divisibility of the nonzero row determinant. We do not replace (27) by a fixed-modulus calibration assertion.

3.1 Exact zero coefficient of the actual packet

Let F_{α_1, α_2} be the row-dependent smooth orbit amplitude before Mellin separation, and let $\Xi_{\alpha_1, \alpha_2}(j)$ be its bounded real periodic factor of period $H_0 = |h|$. Define

$$\vartheta_{H_0}(\alpha_1, \alpha_2) = \frac{1}{H_0} \sum_{r \bmod H_0} \mathbf{1}_{(r, H)=1} \Xi_{\alpha_1, \alpha_2}(r). \quad (28)$$

In the opened TPC-18 square one has, in particular, $\Xi_{\alpha_1, \alpha_2}(r) = \xi(d_1, r)\xi(d_2, r)$. For a Hermitian convention the second factor is conjugated; the absolute estimate below is unchanged. A general fixed finite family is handled cell by cell. In every case

$$|\vartheta_{H_0}(\alpha_1, \alpha_2)| \ll_h 1. \quad (29)$$

Poisson summation on each progression modulo $H_0[u, v]$ shows that the physical zero contribution of one opened dyadic packet is exactly

$$\mathcal{Z}_{S,T}^0 = J \sum_{\alpha_1, \alpha_2} \gamma_{\alpha_1}^{(1)} \gamma_{\alpha_2}^{(2)} \mathbf{m}(\alpha_1, \alpha_2) \vartheta_{H_0}(\alpha_1, \alpha_2) \widehat{F}_{\alpha_1, \alpha_2}(0) K_{S,T}^0(m_{\alpha_1}, m_{\alpha_2}). \quad (30)$$

On the primitive row support $(d_i, H) = 1$, the pure primitive mask gives the local factor $\varphi(H_0)/H_0 = \varphi(H)/H$, recovering the normalization in TPC-24. Formula (30) is taken before the smooth row-orbit weight is decomposed. This is legitimate because evaluation at Fourier frequency zero is the bounded linear functional $F \mapsto \widehat{F}(0) = \int F$.

Theorem 3.1 (Zero before separation). *Assume $R \leq S < T \leq U$, $T \leq X^{C_T}$, the actual opened-row provenance (7)–(11), the bounded off-diagonal mask (12), and the pointwise shell estimate (27). Then, for every fixed $A > 0$ and $0 < \kappa < s$,*

$$|\mathcal{Z}_{S,T}^0| \ll_{A, \kappa, C_T, h} JQ^2 \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} \right\}. \quad (31)$$

Since $QJ \asymp X$, this is

$$|\mathcal{Z}_{S,T}^0| \ll XQ \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} \right\}. \quad (32)$$

In particular, for every fixed $B > 0$,

$$|\mathcal{Z}_{S,T}^0| \ll_{B, C_T, h} XQ (\log X)^{-B}. \quad (33)$$

No fixed-projective-mass hypothesis is required in addition to the actual TPC-18 opened-row provenance.

Proof. By (19) and (29), the scalar multiplying $\gamma_{\alpha_1}^{(1)} \gamma_{\alpha_2}^{(2)} \mathbf{m}(\alpha_1, \alpha_2)$ in (30), apart from the zero kernel, is uniformly bounded. Hence corollary 2.3 gives

$$\sum_{\alpha_1, \alpha_2} \left| \gamma_{\alpha_1}^{(1)} \gamma_{\alpha_2}^{(2)} \mathbf{m}(\alpha_1, \alpha_2) \vartheta_{H_0}(\alpha_1, \alpha_2) \widehat{F}_{\alpha_1, \alpha_2}(0) \right| \ll_h Q^2. \quad (34)$$

Insert (27) into (30) and use (34). This proves (31); (32) follows from $QJ \asymp X$. For (33), first fix, for example, $\kappa = s/2$. The second term then has a fixed power saving and is smaller than every prescribed negative power of $\log X$. Choose the calibration exponent A after B . $\square$

Remark 3.2 (Why the order removes the apparent X^ε). If one first truncates a generic Mellin superposition at X^ε , counts all separated terms, and only then takes absolute values at frequency zero, the coarse bookkeeping can produce $Q^2 X^\varepsilon$. Multiplication by an arbitrarily logarithmic pointwise error does not close that expression. In (30), by contrast, the zero functional is applied directly to the uniformly smooth row-dependent amplitude; then Chebyshev summation gives the exact scale Q^2 . No cancellation between rows is asserted or needed.

3.2 Polylogarithmic reassembly

The conclusion survives the finite bookkeeping used to return from an opened slice. Suppose the original packet is the sum of dyadic, content, sign, and fixed-residue cells $\mathfrak{b}$, with total complexity $O((\log X)^{C_{\text{cell}}})$. Let the scale in cell $\mathfrak{b}$ be $(Q_{\mathfrak{b}}, J_{\mathfrak{b}})$, with $Q_{\mathfrak{b}}J_{\mathfrak{b}} \asymp X$, and suppose the bounded partition coefficients have total variation $O((\log X)^{C_{\text{cell}}})$. Summing (31) gives, for every $B > 0$,

$$\sum_{\mathfrak{b}} |\mathcal{Z}_{S,T}^0(\mathfrak{b})| \ll_B (\log X)^{-B} \sum_{\mathfrak{b}} XQ_{\mathfrak{b}}, \quad (35)$$

after choosing the calibration exponent beyond $B + C_{\text{cell}}$. Equivalently, comparison with the largest cell costs only another fixed power of $\log X$. Prime-power removal and the degenerate row channels retain their previously proved fixed-power margins and are not part of the logarithmic zero-mode ledger.

3.3 Exact scope

The theorem concerns the actual opened TPC packet, not every object in the later abstract interface. In particular:

- (i) An arbitrary matrix z_{α_1, α_2} satisfying only $|z_{\alpha_1, \alpha_2}| \ll X^\varepsilon$ need not obey (17). The conclusion cannot be transferred to such a matrix by deleting its provenance.
- (ii) An arbitrary uniformly smooth family F_{α_1, α_2} does not by itself force the pair coefficient to be a prime–Möbius rank-one weight. What is used is the combination of the actual row factors (10), the bounded zero functional (19), and $|\mathfrak{m}| \leq 1$.
- (iii) The argument closes the physical zero frequency. The nonzero conductor cells still use the TPC-24 large-sieve, no-wrap, and Fourier tail interfaces. Their fixed-power margin is compatible with the conventional X^ε smooth-separation loss.
- (iv) Nothing here estimates the both-new square, the remaining complementary residual legs, or a prime-pair main term. In particular, it is not a Hardy–Littlewood asymptotic, a twin-prime lower bound, or a claim to cross the general sieve parity barrier.

Thus the fixed-projective-mass condition in the abstract TPC-24 corollary remains a valid sufficient hypothesis for arbitrary separated inputs, while [theorem 3.1](#) proves that it is automatic—and can in fact be bypassed—for the physical zero mode of the packet that actually arises from the opened- d construction.

4 Closure of the actual one-sided mixed increment

We now splice the physical zero-frequency estimate into the nonzero exact-conductor result of TPC-24. This section introduces no new conductor estimate. Its role is to state precisely which former hypothesis has been discharged and which inherited interfaces remain.

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad S = X^{s+o(1)}, \quad T = X^{t+o(1)}, \quad (36)$$

where $0 < s \leq t$, $QJ \asymp X$, and put

$$\eta_{\text{asym}}(\beta; s, t) := \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 2s - 2t}{2}, \frac{3 - \beta - s - 5t}{4} \right\}. \quad (37)$$

The three entries are respectively the two-low-conductor, two-high-conductor, and mixed-conductor faces of the asymmetric rectangle.

Let $P_{m,W}$ denote the calibrated prefix of TPC-24 and set

$$L_{m;S,T} = P_{m,T} - P_{m,S}.$$

For one actual primitive opened- d packet, with its original smooth row-orbit weight retained until after physical frequency zero is extracted, let $\Xi_{\alpha,\gamma}(j)$ be its bounded periodic factor of fixed period $H_0 = |h|$, and write

$$\mathcal{I}_{S,T}^{\text{phys}} = \sum_{\alpha,\gamma} \gamma_{\alpha}^{(1)} \gamma_{\gamma}^{(2)} \mathbf{m}(\alpha, \gamma) \sum_{\substack{j \in \mathbb{Z} \\ (j,H)=1}} \Xi_{\alpha,\gamma}(j) \mathcal{W}_{\alpha,\gamma}(j/J) P_{m_{\alpha},S}(j) L_{m_{\gamma};S,T}(j). \quad (38)$$

Here $\gamma^{(1)}, \gamma^{(2)}$, the rows, the mask, and the uniformly smooth amplitude are exactly those of the factorable provenance class in [section 2](#). Equivalently, one may first split the $O_h(1)$ classes modulo H_0 .

Lemma 4.1 (Transfer to the separated nonzero interface). *The continuous Mellin representation (22) transfers (38), restricted to nonzero additive frequencies, to the separated packet interface of TPC-24. More precisely:*

- (i) *every Mellin component satisfies $\|x\|_{\infty} + \|y\|_{\infty} \ll \log(2L)$ and $\|x\|_2^2 + \|y\|_2^2 \ll Q \log(2L)$;*
- (ii) *the actual generic mask is unchanged, remains outside the Mellin integral, and is independent of every divisor and conductor variable;*
- (iii) *every fixed orbit seminorm grows at most polynomially in the Mellin variables, and its weighted total variation is finite; and*
- (iv) *the TPC-24 discrepancy, principal-mean, and Fourier-tail bounds may be integrated componentwise and restored by dominated convergence.*

The row scales and hence the no-wrap inequalities are unchanged.

Proof. Part (i) is [lemma 2.2](#); the Mellin row phases have modulus one. Part (ii) holds because smooth separation acts only on the original row-orbit amplitude. Differentiating the orbit phases introduces a polynomial in τ , so (iii) follows from the Schwartz moments in [equations \(23\) and \(24\)](#). The underlying nonzero-frequency packet is linear. After the triangle inequality and Tonelli's theorem are applied to the Mellin representation, the reassembled Fourier tail is bounded by the integral of the component tail masses. The same Schwartz moments dominate that integral and the polynomial constants in the inherited nonzero packet estimates. Finite Riemann sums therefore obey those estimates uniformly, and dominated convergence proves (iv). Splitting the fixed H_0 -periodic factor costs only $O_h(1)$. $\square$

Let $\mathcal{D}$ denote the fixed data consisting of h, δ, C_T , the positive row, no-wrap, Fourier-tail, and conductor margins, and the fixed smooth and residue families.

Theorem 4.2 (Actual one-sided closure). *Assume the primitive row and target-coprimality conditions of TPC-24, its divisor-independent off-diagonal generic mask, its no-wrap and polynomial Fourier-tail interfaces, and $L \geq X^{\lambda_0}$ for a fixed $\lambda_0 > 0$. Suppose*

$$R \leq S < T \leq U, \quad T \leq X^{C_T}, \quad (39)$$

and

$$\eta_{\text{asym}}(\beta; s, t) > 0 \quad (40)$$

with a fixed margin. Then for every fixed $B > 0$,

$$\boxed{\mathcal{I}_{S,T}^{\text{phys}} \ll_{B,\mathcal{D}} X Q(\log X)^{-B}.} \quad (41)$$

The same estimate holds for the transpose packet and for a general nonzero-mean fixed smooth test within this factorable provenance class. No separate fixed-projective-mass hypothesis and no mean-zero hypothesis on the fixed physical test are required.

Proof. The exact calibrated-prefix identity of TPC-24 writes the product in (38) as the joint short-long divisor lattice minus the explicit row drift. Treat the joint physical zero frequency before smooth separation. The zero-before-separation theorem in section 3 bounds it by

$$XQ \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} \right\} \quad (42)$$

for every fixed $A > 0$ and $0 < \kappa < s$. Fixed bounded-overlap reassembly costs only $(\log X)^{O(1)}$, so A may be chosen after B . The second term has a fixed power saving.

For nonzero additive frequencies, apply lemma 4.1 and the asymmetric exact-conductor estimate of TPC-24 [8]. Positivity of (37) with fixed margin gives a fixed-power saving after all dyadic and divisor multiplicities. The global principal conductor cell is $O(Q^2 X^\varepsilon)$, hence has a fixed-power saving relative to $Q^2 J = XQ$ because $J = X^{1-\beta+o(1)}$ has a fixed positive exponent in the stated region. The polynomial Fourier tail is made smaller than any prescribed fixed power by choosing its integration-by-parts order after its polynomial mass.

Finally, the calibrated row drift contributes $O(JQ^2 L^{-1} X^\varepsilon)$. Choose the auxiliary ε smaller than the fixed L -margin. Every term other than (42) then has a fixed-power saving, and the calibration exponent in that display supplies the requested arbitrary logarithmic saving. $\square$

Remark 4.3 (What was removed). TPC-24 proves the same conclusion for an abstract separated packet under the additional bound $\sum_\nu \|x^{(\nu)}\|_1 \|y^{(\nu)}\|_1 \ll Q^2 (\log X)^K$. Theorem 4.2 removes that as an *independent* hypothesis only for the actual opened- d provenance class. The no-wrap, polynomial Fourier-tail, row geometry, and positive conductor-margin assumptions have not been removed.

4.1 The strict crossing window

For a genuine crossing of the old Poisson face, impose

$$s < 1 - \beta < t. \quad (43)$$

The positivity conditions in (37) give

$$1 - \beta < t < \min \left\{ \frac{\beta + 1}{2} - s, \frac{3 - \beta - s}{5} \right\}, \quad (44)$$

together with $\beta < 2/3$ and $s < 1 - \beta$, all with fixed margins.

At the baseline cutoff

$$S = R = X^{1/2-\delta+o(1)}, \quad T = RX^{\xi+o(1)}, \quad (45)$$

the saving becomes

$$\eta_{\text{asym}}(\beta, \delta, \xi) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 4\delta - 1}{2} - \xi, \frac{6\delta - \beta - 5\xi}{4} \right\}. \quad (46)$$

The strict passage $R < J < T$ is certified when

$$\boxed{\frac{1}{2} + \delta - \beta < \xi < \min \left\{ \frac{\beta + 4\delta - 1}{2}, \frac{6\delta - \beta}{5} \right\}}, \quad (47)$$

with $\beta < 2/3$ and $\beta < 1/2 + \delta$. For example,

$$\delta = \frac{3}{20}, \quad \beta = \frac{31}{50}, \quad \xi = \frac{1}{25} \quad (48)$$

gives $R = X^{7/20}$, $J = X^{19/50}$, $T = X^{39/100} > J$, and nonzero-conductor saving $1/50$. The new content is not a larger exponent window; it is that the physical packet now satisfies the general-mean conclusion throughout that window without an unverified projective-mass assumption.

5 The principal row mode and the energy boundary

This section records a sharp limitation of the soft row interface. The physical zero kernel may be very small at every individual row pair and still possess a coherent row direction whose singular value is the row cardinality times the pointwise size. Consequently, the soft $QX^{o(1)}$ row-energy interface, together with the Hilbert–Schmidt bound obtained only from entrywise control, cannot replace the absolute row-pair mass. The obstruction already occurs in the shared-factor block $g = 1$, where compatibility supplies no oscillation in the row variables.

5.1 An abstract soft-energy saturation theorem

Let $Q = X^{\beta+o(1)}$ for some fixed $\beta > 0$, and let $\mathcal{R}_X$ be a row set of cardinality

$$N = N_X = \left\lfloor \frac{Q}{\log(2Q)} \right\rfloor. \quad (49)$$

Write $\mathbf{1}$ for the constant vector on $\mathcal{R}_X$, and put

$$\rho_X = \exp\{-\sqrt{\log X}\}. \quad (50)$$

Thus $\rho_X^{-1} = X^{o(1)}$, while for every fixed $A > 0$,

$$\rho_X \ll_A (\log X)^{-A}. \quad (51)$$

Theorem 5.1 (Soft ℓ^2 energy does not close a dense zero mode). *There are a divisor-independent off-diagonal row mask, a real kernel, and row weights satisfying*

$$\|x\|_\infty + \|y\|_\infty \ll_\epsilon X^\epsilon, \quad \|x\|_2^2 + \|y\|_2^2 \ll_\epsilon QX^\epsilon \quad (52)$$

for every fixed $\epsilon > 0$, such that the kernel obeys the arbitrary-logarithmic pointwise bound

$$\sup_{m,n \in \mathcal{R}_X} |K_X(m,n)| \ll_A (\log X)^{-A} \quad (53)$$

for every fixed $A > 0$, but

$$\sum_{m,n \in \mathcal{R}_X} x_m y_n \mathbf{m}_X(m,n) K_X(m,n) = Q^2 \{1 + o(1)\}. \quad (54)$$

In particular, the three properties in (52)–(53) do not imply either a fixed-power saving or an arbitrary-logarithmic saving from the natural row scale Q^2 .

Proof. Take

$$\mathbf{m}_X(m,n) = \mathbf{1}_{\{m \neq n\}}, \quad K_X(m,n) = \rho_X, \quad x_m = y_m = \frac{Q}{N\sqrt{\rho_X}}. \quad (55)$$

The mask is bounded, off-diagonal, and independent of every divisor variable. Equations (50) and (49) give

$$\|x\|_\infty = \|y\|_\infty = \frac{Q}{N\sqrt{\rho_X}} = X^{o(1)}, \quad (56)$$

$$\|x\|_2^2 + \|y\|_2^2 = \frac{2Q^2}{N\rho_X} = QX^{o(1)}. \quad (57)$$

This proves (52), while (53) is (51). Finally,

$$\sum_{m,n} x_m y_n \mathbf{m}_X(m,n) K_X(m,n) = \frac{Q^2}{N^2 \rho_X} \rho_X N(N-1) = Q^2 \left(1 - \frac{1}{N}\right),$$

which proves (54). $\square$

For any fixed finite divisor set, or in the prime-row regime in which all rows exceed the divisor horizon, the construction can be placed on primitive rows coprime to every divisor variable under consideration. In the $g = 1$ block the compatibility condition is automatic, so target primitivity merely deletes inadmissible one-row factors; it does not create cancellation between different rows. The theorem is intentionally an interface countermodel: K_X is not asserted to be the actual Möbius shell kernel.

5.2 Operator norm, Hilbert–Schmidt norm, and stable rank

The masked matrix in the preceding proof is

$$A_X = \rho_X(\mathbf{1}\mathbf{1}^\top - I_N). \quad (58)$$

The constant vector has eigenvalue $\rho_X(N-1)$, while its orthogonal complement has eigenvalue $-\rho_X$. Hence

$$\|A_X\|_{2 \rightarrow 2} = \rho_X(N-1), \quad (59)$$

$$\|A_X\|_{\text{HS}} = \rho_X \sqrt{N(N-1)}, \quad (60)$$

$$\text{srnk}(A_X) := \frac{\|A_X\|_{\text{HS}}^2}{\|A_X\|_{2 \rightarrow 2}^2} = \frac{N}{N-1} \rightarrow 1. \quad (61)$$

Thus the Hilbert–Schmidt norm has the same order as the operator norm. There is no square-root row gain to extract: after diagonal removal the dangerous matrix is rank one plus a scalar diagonal correction.

More generally, for arbitrary row weights,

$$x^\top(\mathbf{1}\mathbf{1}^\top - I_N)y = \left(\sum_m x_m\right)\left(\sum_n y_n\right) - \sum_m x_m y_m. \quad (62)$$

The off-diagonal mask removes only the second, diagonal expression. It does not center either row sequence against the constant vector.

5.3 A finite exact Möbius certificate

The coherent geometry above is present in the exact divisor algebra, not only in the abstract countermodel. Recall

$$a(w) = -\mu(w) \log w, \quad \lambda'_R(w) = w \sum_{b \leq R/w} \frac{\mu(wb)\mu(b)}{\varphi(wb)}, \quad b_R(w) = a(w) - \lambda'_R(w). \quad (63)$$

Proposition 5.2 (An exact constant zero kernel). *Take*

$$H = 1, \quad R = S = 2, \quad T = 3, \quad (64)$$

and let the two row variables range over any finite set $\mathcal{R}$ of at least two primes greater than 3. Then for every distinct pair $m, n \in \mathcal{R}$, the full physical zero kernel equals

$$K_{2,3}^0(m, n) = \kappa_0, \quad \boxed{\kappa_0 = \frac{(\log 2 - 2) \log 3}{6} \neq 0.} \quad (65)$$

Consequently, after applying $\mathbf{m}(m, n) = \mathbf{1}_{\{m \neq n\}}$, its matrix is exactly

$$\kappa_0(\mathbf{1}\mathbf{1}^\top - I_N). \quad (66)$$

Its operator norm, Hilbert–Schmidt norm, and stable rank are obtained from (59)–(61) by replacing ρ_X with $|\kappa_0|$.

Proof. Here $u \in \{1, 2\}$ and the long shell contains only $v = 3$. Thus $(u, v) = 1$ for every effective divisor pair. Since the rows are primes greater than 3, all target-primitivity restrictions are automatic, and the compatibility indicator is identically one. Directly from (63),

$$\begin{aligned} \lambda'_2(1) &= 2, & \lambda'_2(2) &= -2, \\ b_2(1) &= -2, & b_2(2) &= \log 2 + 2, & a(3) &= \log 3. \end{aligned}$$

Therefore

$$\begin{aligned} K_{2,3}^0(m, n) &= \frac{a(3)}{3} \left(b_2(1) + \frac{b_2(2)}{2} \right) \\ &= \frac{\log 3}{3} \left(-2 + \frac{\log 2 + 2}{2} \right) = \frac{(\log 2 - 2) \log 3}{6}. \end{aligned}$$

The constant is negative, hence nonzero, because $\log 2 < 2$. The matrix assertion follows by deleting its diagonal. $\square$

Remark 5.3 (What the finite certificate proves). The proposition proves that the principal row direction is compatible with the genuine Möbius coefficients, exact target primitivity, and exact shared-factor compatibility. It also proves that diagonal deletion is not row centering. It does not provide an asymptotic lower bound for a growing wide shell and therefore is not a counterexample to an eventual power-saving estimate for the actual TPC shell.

5.4 Interface obstruction versus number-theoretic obstruction

The preceding results have two deliberately different logical levels.

- (i) [Theorem 5.1](#) is a black-box non-implication theorem. It proves that soft $X^{a(1)}$ row energy, pointwise arbitrary-logarithmic decay, divisor independence of the mask, and target primitivity do not by themselves imply a useful bilinear estimate.
- (ii) [Proposition 5.2](#) is exact arithmetic. It proves that the coherent spectral geometry occurs in an actual finite Möbius shell.
- (iii) Neither result says that the zero kernel in the intended growing shell is large. A genuine asymptotic counterexample would require a quantitative lower bound along those growing shells and row weights that arise in the original decomposition. No such lower bound is asserted here.

There is a second black-box obstruction if a theorem is stated for every bounded row-pair mask. Given any real off-diagonal matrix $K(m, n)$, the row-only choice $\mathbf{m}(m, n) = \text{sgn } K(m, n)$ converts its bilinear sum with positive constant weights into $\sum_{m \neq n} |K(m, n)|$. It is still independent of the individual divisor variables. Therefore a positive row-cancellation theorem must either retain and analyze the actual generic mask or impose a quantitative structural norm on the class of masks; the entrywise bound $|\mathbf{m}| \leq 1$ is not enough.

5.5 Minimal additional inputs

The obstruction isolates three viable forms of extra information.

Principal-row balance. For the exact model $\kappa(\mathbf{1}\mathbf{1}^\top - I_N)$, (62) shows that the minimal new datum is cancellation of at least one row sequence against the constant vector. A quantitative form is

$$\frac{|\langle x, \mathbf{1} \rangle|}{\sqrt{N}\|x\|_2} \frac{|\langle y, \mathbf{1} \rangle|}{\sqrt{N}\|y\|_2} \ll X^{-\sigma} \quad (67)$$

for some fixed $\sigma > 0$; exact mean zero on either side is a stronger sufficient condition. The remaining diagonal term is bounded by $|\kappa|\|x\|_2\|y\|_2$, with no factor N .

For a shared factor $g > 1$, the analogous coherent subspace is spanned by the indicators of the admissible residue classes modulo g , with the local coprimality multipliers included. Thus a general positive theorem needs a uniform bound for the projections of the actual row weights onto these residue-class constant spaces. Additive large-sieve energy may then be applied to the complementary, nonprincipal row frequencies. It cannot treat $g = 1$, because that block has no nonprincipal row frequency.

Structured-mask control. Principal balance must be preserved after inserting the actual mask. One sufficient interface is a separated representation

$$\mathbf{m}(m, n) = \sum_{\nu} \alpha_{\nu}(m) \beta_{\nu}(n), \quad \sum_{\nu} \|\alpha_{\nu}\|_{\infty} \|\beta_{\nu}\|_{\infty} \ll (\log X)^C, \quad (68)$$

together with (67) for every twisted pair $x\alpha_{\nu}, y\beta_{\nu}$. Alternatively one may work directly with the specific generic mask and prove that its exceptional part has a power-saving operator norm. A generic bounded Schur mask cannot simply be discarded before taking the nonlinear norm.

Fixed-polylogarithmic row energy. There is also a simpler sufficient hypothesis which does not exploit row cancellation. If $\#\mathcal{R}_X \ll Q$ and

$$\|x\|_2 \|y\|_2 \ll Q(\log X)^C, \quad (69)$$

then a pointwise bound $\sup |K| \ll_A (\log X)^{-A}$ gives

$$|x^\top (\mathbf{m} \circ K) y| \leq \|x\|_2 \|y\|_2 \|\mathbf{m} \circ K\|_{2 \rightarrow 2} \ll_A Q^2 (\log X)^{C-A}.$$

Choosing A after the desired logarithmic saving closes the term. For a projective sum of row packets, the corresponding sum of the ℓ^2 -norm products must satisfy (69). This route is rigorous but is a fixed-polylogarithmic mass input in another norm; it is not a consequence of the soft QX^c energy estimate.

Absent a fixed-polylogarithmic mass input, a genuinely cancellation-based route is therefore not “apply the entrywise Hilbert–Schmidt bound once more.” It must isolate the principal row projections, prove a balance statement for the actual weights and mask, and only then use large-sieve energy on the complementary row spectrum.

6 The congruence skeleton and a deflated row estimate

The principal-mode obstruction identifies what a genuine row-space argument must remove. In this section we make that geometry exact in a prime-row model. The model is deliberately narrower than the physical opened- d packet used in the positive theorem: it has no small row factor and no additional generic Schur pruning. Its purpose is to identify the exceptional row subspaces, not to replace the provenance argument of [section 3](#).

Fix $H \geq 1$, let $R \leq S < T < L/2$, and take a finite set

$$\mathcal{P}_L \subset \{p \in [L/2, 2L] : p \text{ is prime, } (p, H) = 1\}. \quad (70)$$

In particular every row prime is larger than every divisor variable in the shell. For $1 \leq g \leq S$, define

$$\omega_g := \frac{1}{g} \sum_{\substack{c \leq S/g \\ (gc, H) = 1}} \frac{b_R(gc)}{c} \sum_{\substack{S/g < d \leq T/g \\ (d, gcH) = 1}} \frac{a(gd)}{d}, \quad (71)$$

with the convention that ineffective nonsquarefree terms vanish. Also let A_g be the congruence matrix on $\ell^2(\mathcal{P}_L)$,

$$A_g(m, n) = \mathbf{1}_{m \equiv n \pmod{g}}. \quad (72)$$

Theorem 6.1 (Exact prime-row congruence skeleton). *Let $\mathbf{K}_{S,T}^0$ be the matrix whose off-diagonal entries are the physical zero kernel [\(26\)](#) and whose diagonal entries are zero. Then*

$$\boxed{\mathbf{K}_{S,T}^0 = \sum_{g \leq S} \omega_g (A_g - I)}. \quad (73)$$

For

$$v_{g,r}(m) = \mathbf{1}_{m \equiv r \pmod{g}}, \quad r \bmod g, \quad (74)$$

one has

$$A_g = \sum_{r \bmod g} v_{g,r} v_{g,r}^*, \quad \text{rank } A_g = \#\{m \bmod g : m \in \mathcal{P}_L\} \leq g. \quad (75)$$

Moreover,

$$\|A_g\|_{2 \rightarrow 2} = \max_{r \bmod g} \#\{m \in \mathcal{P}_L : m \equiv r \pmod{g}\} \ll \frac{L}{g}. \quad (76)$$

Thus the correlated part of every g -layer lies exactly in the residue-class constant space modulo g .

Proof. Because every row prime exceeds T , the row coprimality conditions in [\(26\)](#) reduce to coprimality with H . Put $g = (u, v)$, $u = gc$, and $v = gd$. Squarefree support makes g, c, d pairwise coprime on every effective term. The exact gcd reparametrization becomes, for $m \neq n$,

$$K_{S,T}^0(m, n) = \sum_{\substack{g \leq S \\ g | m-n}} \omega_g.$$

The matrix $A_g - I$ has precisely the off-diagonal entries $\mathbf{1}_{g | m-n}$ and zero diagonal, proving [\(73\)](#).

The first identity in [\(75\)](#) follows by partitioning the rows by their residue class. After this partition, A_g is a direct sum of all-ones blocks, one for each occupied class. Each nonempty block has rank one and its only nonzero eigenvalue is its cardinality. This proves the rank formula and the first equality in [\(76\)](#). An interval of length $O(L)$ contains $O(L/g + 1) = O(L/g)$ integers in one class because $g \leq S < L/2$, proving the final estimate. $\square$

The scalar diagonal in (73) is harmless at the row-pair scale, but it must not be silently dropped. Put

$$\Omega_{S,T} = \sum_{g \leq S} \omega_g, \quad \mathcal{V}_G = \text{span}\{v_{g,r} : 1 \leq g \leq G, r \bmod g\}. \quad (77)$$

Then

$$\mathbf{K}_{S,T}^0 = \sum_{g \leq S} \omega_g A_g - \Omega_{S,T} I.$$

Lemma 6.2 (Coefficient mass of one skeleton layer). *Uniformly for $g \leq S$,*

$$|\omega_g| \ll_{C_{T,H}} \frac{(\log(2X))^6}{g}, \quad |\Omega_{S,T}| \ll_{C_{T,H}} (\log(2X))^7, \quad (78)$$

provided $T \leq X^{C_T}$.

Proof. The prefix cofactor estimate of TPC-24 gives

$$\sum_{c \leq S/g} \frac{|b_R(gc)|}{c} \ll (\log(2X))^4,$$

uniformly in squarefree g . The corresponding harmonic bound for the long coefficient is

$$\sum_{d \leq T/g} \frac{|a(gd)|}{d} \ll (\log(2X))^2.$$

Insert these estimates into (71). Summing $1/g$ for $g \leq S$ proves the second assertion. $\square$

Theorem 6.3 (Deflated operator estimate). *Let $1 \leq G < S$. If $x \in \ell^2(\mathcal{P}_L)$ satisfies*

$$x \perp \mathcal{V}_G, \quad (79)$$

then for every $y \in \ell^2(\mathcal{P}_L)$,

$$\boxed{|\langle x, \mathbf{K}_{S,T}^0 y \rangle| \ll_{C_{T,H}} \left\{ \frac{L(\log(2X))^6}{G} + (\log(2X))^7 \right\} \|x\|_2 \|y\|_2.} \quad (80)$$

The same conclusion holds if the balance condition is imposed on y instead.

Proof. For $g \leq G$, the range of A_g is contained in $\mathcal{V}_G$, so $\langle x, A_g y \rangle = 0$. By theorem 6.1 and lemma 6.2,

$$\begin{aligned} |\langle x, \mathbf{K}_{S,T}^0 y \rangle| &\leq \left(\sum_{g > G} |\omega_g| \|A_g\|_{2 \rightarrow 2} + |\Omega_{S,T}| \right) \|x\|_2 \|y\|_2 \\ &\ll \left(L(\log(2X))^6 \sum_{g > G} \frac{1}{g^2} + (\log(2X))^7 \right) \|x\|_2 \|y\|_2, \end{aligned}$$

which is (80). Symmetry gives the last statement. $\square$

For instance, if $L = X^{\lambda+o(1)}$, $G = X^{\gamma+o(1)}$ with $0 < \gamma < \min\{s, \lambda/2\}$, and $\|x\|_2 \|y\|_2 \ll LX^\varepsilon$, then (80) is

$$\ll L^2 X^{-\gamma+\varepsilon+o(1)} + LX^{\varepsilon+o(1)}, \quad (81)$$

a fixed-power saving from the prime-row pair scale L^2 after choosing ε below the fixed margins. This is a genuine row-averaged theorem, but its balance hypothesis is strong and explicit: since

$v_{1,0} = \mathbf{1}$, it first removes the principal row mode, and it then removes every residue-class constant mode up to G .

The balance space is nonvacuous for the full dyadic prime-row family: indeed $\dim \mathcal{V}_G \leq \sum_{g \leq G} g \ll G^2$, while the prime number theorem gives $\#\mathcal{P}_L \asymp L/\log L$. Thus the displayed strict condition $2\gamma < \lambda$ leaves a nonzero orthogonal complement for all sufficiently large X . For an arbitrary sparse subset $\mathcal{P}_L$, (80) remains valid but the existence of a nonzero balanced vector must be checked separately.

Remark 6.4 (Why this is not the physical closure theorem). The actual opened row is $m = \ell d_0$. Local coprimality inserts diagonal row multipliers, and for fixed d_0 the exceptional spaces become residue-class constant spaces inside each d_0 -block. A global mean-zero condition need not center every such block. Moreover, the near-determinant and row-gcd conditions form a Schur mask; entrywise masking does not preserve the exact ranges in (75). One must either enlarge $\mathcal{V}_G$ to include those blockwise multipliers and control the mask as a sparse operator perturbation, or use the zero-before-separation absolute-mass route. The present paper takes the latter route for the one-sided physical packet. No deflated estimate for the both-new physical square is claimed here.

7 Exact certificate, scope, and conclusion

The analytic input in this paper is symbolic: Chebyshev’s upper bound, the TPC-24 endpoint calibration theorem, and the previously established nonzero-conductor estimates. A deterministic finite certificate is included to audit the exact divisor and row-space identities that are most vulnerable to normalization, diagonal, or shared-factor errors.

7.1 Certificate

The standard-library script

`experiments/tpc25_certificate.py`

uses `fractions.Fraction` throughout. Logarithms are represented as formal prime variables, so no floating-point comparison enters the kernel tests. It verifies:

- (1) the direct physical zero kernel against its exact congruence- skeleton expansion on three finite prime-row families, including nontrivial fixed target radicals;
- (2) the identity $K_{2,3}^0 = ((\log 2 - 2) \log 3)/6$ on every off-diagonal entry of a five-row model;
- (3) the principal and contrast eigenvector relations for the exact rank-one-plus-diagonal matrix;
- (4) the rank of each congruence matrix $A_g = (\mathbf{1}_{m \equiv n \pmod{g}})$, compared with the number of occupied residue classes and the upper bound $\text{rank } A_g \leq g$;
- (5) a rational finite saturation model for the energy-only counterexample; and
- (6) the rational crossing sample and all three entries of the asymmetric exponent ledger.

Independent ordinary and `python -O` runs perform the same exact checks and are required to generate byte-identical JSON. The archived JSON records false flags for asymptotic Möbius evidence, full residual closure, Hardy–Littlewood, twin primes, and a general parity breakthrough. Hashes are recorded in the repository README after the final source audit.

The certificate does not test Chebyshev’s theorem, the classical zero-free region, or any asymptotic cancellation statement. Its role is regression control for finite algebra, not numerical evidence.

7.2 What is unconditional and what is inherited

Within one actual opened- d packet, the following statements are finite or elementary and unconditional:

- (i) the row-provenance factorization and the $O(Q)$ one-row absolute mass;
- (ii) extraction of physical frequency zero before smooth separation;
- (iii) the resulting $O(Q^2)$ zero-frequency pair mass;
- (iv) the soft-energy countermodel and the finite exact Möbius principal mode; and
- (v) the congruence-skeleton and deflated operator identities in the stated prime-row model.

The arbitrary-logarithmic pointwise shell bound is imported from TPC-24; it ultimately uses the classical zero-free region for $\zeta(s)$ in a uniform Möbius calibration [1, 8]. The nonzero-frequency closure also imports the TPC-24 no-wrap, polynomial Fourier-tail, and exact-conductor hypotheses. The present paper removes only the independent fixed-projective-mass condition for the actual physical packet.

The operator obstruction must be read with equal precision. It proves that the abstract soft interfaces do not logically imply a row-averaged saving and that the coherent direction occurs in exact finite Möbius algebra. It does not prove a lower bound for the intended growing shell, and therefore it does not show that a deeper arithmetic row- average theorem is false.

7.3 Conclusion and next gate

TPC-24 left the physical zero mode at a clean but mass-sensitive boundary. Returning to the pre-separated packet resolves that boundary for the actual one-sided opened- d correlation. A direct provenance proof is obtained in the following order:

extract physical zero frequency $\longrightarrow$ use row provenance $\longrightarrow$ separate only the nonzero spectrum	(82)
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This yields arbitrary logarithmic saving for a general nonzero-mean fixed smooth test in the factorable provenance class throughout the existing positive asymmetric-conductor region, without assuming a fixed projective norm that was not proved in TPC-24.

The energy audit also prevents a false shortcut. Entrywise arbitrary-logarithmic decay and soft L^2 row energy cannot be fed into a black-box Hilbert–Schmidt estimate: the shared-factor block $g = 1$ contains a coherent principal row mode. In the prime-row model the remaining small- g directions are exactly the residue-class constant spaces. A future operator approach must therefore center those projections while keeping the actual generic Schur mask and the local coprimality multipliers inside the argument.

The next analytic component is the both-new square $L_{m;S,T}L_{n;S,T}$. It lacks the calibrated short prefix whose two endpoint constants cancel in the one-sided argument. One possible route is a row-space projection theorem for its low- g congruence skeleton followed by a nonprincipal additive large sieve. A second is to exploit a different exact recombination before either divisor leg is estimated. Failure of either route would still yield a precise operator or arithmetic obstruction; success would close only another residual component.

Even after such a step, the ultra-long complement fibers, global endpoint reassembly, positivity, and the prime-pair main term remain. Accordingly this paper makes no Hardy–Littlewood, twin-prime, or general parity-breaking claim.

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